



Research Report

Project: Surplus Food Redistribution Network

Team: Catalyst Alliance

UN Sustainable Development Goal: Goal 2 – Zero Hunger

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Research Overview

This research explores how surplus food from supermarkets, grocery stores, bakeries, and other retail vendors can be redistributed to NGOs, food banks, community organizations, and vulnerable populations through a digital platform.

Although Africa produces significant amounts of food, millions of people across the continent continue to experience hunger and food insecurity while edible food is lost or wasted throughout the food supply chain. This project proposes a technology-driven solution that connects food donors with organizations that can safely redistribute surplus food before it goes to waste.

The project aligns primarily with **United Nations Sustainable Development Goal (SDG) 2: Zero Hunger**, while also supporting **SDG 12.3**, which aims to reduce food loss and food waste, and **SDG 13: Climate Action**, since reducing food waste also lowers greenhouse gas emissions. ([Uneca](#))

Methodology

The research combined evidence from African and international reports, academic literature, and publicly available datasets. Due to the limited availability of publicly accessible operational retail food waste datasets for African countries, a cleaned, publicly available inventory dataset was analyzed to demonstrate how the proposed redistribution system can identify surplus food, prioritize redistribution opportunities, and support decision-making. Together, the literature review and data analysis informed the design of the proposed solution.

1. The African Context

Food insecurity remains one of Africa's most pressing development challenges. According to the **Africa Regional Overview of Food Security and Nutrition 2025**, Africa is **not on track to achieve SDG 2 (Zero Hunger) by 2030**. The report notes that **more than 300 million people in Africa were undernourished in 2024**, and **nearly 60% of the population experienced moderate or severe food insecurity**. It also highlights that many people cannot afford a healthy diet because of rising food prices, economic pressures, climate change, and conflict. ([Uneca](#))

Implications for the Proposed Solution

These findings show that improving food security in Africa is not only about producing more food. It also requires improving the distribution of existing food. A platform that connects surplus food from retailers with organizations serving people in need can help reduce hunger and improve access to food.

2. Food Waste and Hunger Exist at the Same Time

The United Nations Environment Programme (UNEP) estimates that **1.05 billion tonnes of food were wasted globally in 2022**, even as hundreds of millions of people faced hunger. Food waste occurs in homes, restaurants, retail stores, supermarkets, and other parts of the food supply chain. Many food items discarded by retailers are still safe for consumption but are thrown away because they are close to their best-before date, have damaged packaging, or cannot be sold quickly enough.

This presents an opportunity to recover surplus food and redirect it to communities experiencing food insecurity rather than allowing it to end up in landfills. Food redistribution is therefore recognized as one practical strategy for reducing waste while improving food security. The **UNEP Food Waste Index Report 2024** also highlights that food waste occurs across African countries, including Nigeria, Kenya, and South Africa. While detailed operational retail food waste datasets remain limited, the report reinforces that food waste is a significant regional challenge and supports the need for improved food recovery and redistribution systems. ([UNEP Food Waste Index Report 2024](#))

Implications for the Proposed Solution

Baki focuses specifically on edible surplus food from retailers. Instead of allowing this food to be wasted, Baki makes it visible to verified NGOs and food banks that can collect and redistribute it before it expires.

3. Challenges Facing Food Redistribution in Africa

Although many businesses are willing to donate surplus food, several barriers prevent effective redistribution.

Common challenges include:

- Lack of awareness about nearby organizations that accept food donations.
- Slow communication between food donors and recipient organizations.
- Transportation and logistics difficulties.
- Limited cold-chain infrastructure for perishable food.
- Manual coordination through phone calls or messaging.
- Lack of a centralized platform that connects food donors and recipients.

Research on food accessibility across Africa also shows that geographic inequalities and travel times affect access to food markets, particularly for rural and underserved communities. Digital tools can help improve coordination and identify areas where access is most limited. ([arXiv](#))

Implications for the Proposed Solution

A digital platform such as Baki can reduce these coordination barriers by providing a centralized space where vendors can list surplus food and potential recipients can discover, claim, and coordinate collection. Location-based matching and automated notifications can further reduce the time between a surplus listing and its redistribution.

4. How Baki Can Help

Baki, meaning “*what’s left*” or “*the remainder*” in Swahili, is a lightweight digital surplus food redistribution platform designed to address the coordination gap between food vendors with surplus food and nearby organizations that can redistribute it.

Baki allows vendors to list surplus food by specifying the food type, quantity, expiry date, and pickup location. Vendors can choose to donate their surplus to a pantry or NGO or sell it to the community at a discounted price.

For donations, Baki’s matching engine identifies suitable nearby pantries and NGOs based on proximity, urgency, and available capacity, then sends a notification to the selected organization with the relevant pickup details. The listing also remains visible on a public live board in case the automated match is not collected in time.

For discounted sales, the listing is posted directly to the public live board at a vendor-set price, while community subscribers in the relevant location can receive notifications about new listings.

Retail Stores

- Register as verified food donors.
- Upload available surplus food.
- Specify quantity, collection time, and expiry date.
- Notify nearby organizations instantly.

NGOs and Food Banks

- Register and verify their organization.
- Browse available donations.
- Reserve food.
- Arrange collection.

Platform Features

- Real-time notifications.
- Location-based donor matching.
- Pickup scheduling.
- Donation history.
- Reporting dashboard.
- Impact statistics.

These features are designed to address the coordination barriers identified through the research, reduce delays, improve transparency, and increase the amount of surplus food successfully recovered and redistributed.

5. Existing Platforms We Can Learn From

Several successful food redistribution initiatives already exist around the world.

Too Good To Go helps retailers reduce food waste by making surplus food available before it is discarded.

OLIO connects households and local businesses with nearby people who can use surplus food.

Feeding America works with retailers and manufacturers to recover surplus food through a nationwide network of food banks.

These examples demonstrate that technology can successfully reduce food waste. However, our proposed platform would be adapted to the African context, considering local infrastructure, transportation challenges, and partnerships with NGOs and community organizations.

6. How Our Project Supports the Sustainable Development Goals

Our proposed platform directly contributes to several SDGs.

SDG 2 – Zero Hunger

The platform increases access to nutritious food by connecting surplus food with organizations serving vulnerable communities. ([Uneca](#))

SDG 12.3 – Responsible Consumption and Production

The project helps reduce food waste by ensuring edible food is redistributed rather than discarded. ([World Health Organization](#))

SDG 13 – Climate Action

Reducing food waste also reduces methane emissions from landfills and contributes to more sustainable food systems. ([World Health Organization](#))

7. Data Analysis Findings

The proposed solution was further informed through an analysis of a cleaned, publicly available operational inventory dataset. Although the dataset is not specific to African retailers, it was used to demonstrate how surplus inventory can be identified and prioritized for redistribution in environments where comparable African operational datasets are unavailable.

The analysis revealed several insights relevant to the proposed redistribution network:

- A total of **2,193 overstocked inventory units** were identified, representing products suitable for redistribution before becoming waste.

- **Fruits and Vegetables** recorded the highest surplus inventory and the lowest inventory turnover, making them the highest priority category for redistribution.
- **Dairy products** also showed high surplus levels and slow inventory movement, increasing their risk of spoilage.
- Products such as **Almond Flour, Arabica Coffee, Rice Flour, and Palm Oil** were identified as high-priority redistribution candidates.
- The warehouse located at **30880 Barnett Junction** contained the highest concentration of products approaching expiration.
- Overstock was concentrated among suppliers including **Katz, Skippad, and Youbridge**, suggesting procurement practices could be optimized.
- Inventory turnover analysis highlighted several slow-moving categories requiring closer monitoring, including Fruits and Vegetables, Dairy, Grains and Pulses, Oils and Fats, Bakery, Seafood, and Beverages.

These findings demonstrate how inventory analytics can support proactive surplus food redistribution by identifying where intervention is most needed.

Key Recommendations

Based on this research, our proposed solution should:

- Develop a simple and user-friendly digital platform.
- Focus on supermarkets, grocery stores, bakeries, and fresh produce vendors as food donors.
- Partner with NGOs, food banks, community kitchens, and charities as food recipients.
- Include location-based matching to reduce transportation time.
- Provide real-time notifications for newly available surplus food.
- Track impact through dashboards showing food saved, donations completed, and estimated meals provided.
- Build partnerships with governments, NGOs, and private businesses to encourage long-term adoption.

Conclusion

Our proposed solution, Baki, is a digital surplus food redistribution platform designed to bridge the gap between retailers with surplus food and organizations serving communities in need. Food insecurity and food waste remain major challenges across Africa. While millions of people struggle to access nutritious food, significant quantities of edible food continue to be discarded. Research

shows that improving food redistribution systems can help reduce waste while strengthening food security.

By improving coordination, increasing transparency, and reducing delays, Baki supports the achievement of SDG 2 (Zero Hunger) and contributes to a more sustainable and equitable food system across Africa. Its design is informed by the research findings, particularly the need for faster, hyper-local coordination between food donors and organizations or community members able to use surplus food.

Limitations

Although the project is grounded in African reports, statistics, and literature, publicly available operational retail food waste datasets for African countries are limited. Consequently, a cleaned publicly available operational inventory dataset was used to demonstrate how the proposed redistribution system functions. The analytical approach remains applicable to African contexts, while acknowledging that future implementations should incorporate country-specific operational data where available.

References

1. United Nations Economic Commission for Africa. *Africa Regional Overview of Food Security and Nutrition 2025*. ([Uneca](#))
2. United Nations Economic Commission for Africa. *Sustainable Development Goal 2 (Zero Hunger)*. ([Uneca](#))
3. United Nations Statistics Division. *Sustainable Development Goals Report – Goal 2*. ([UNSD](#))
4. World Health Organization, FAO, IFAD, UNICEF, WFP & WHO. *The State of Food Security and Nutrition in the World 2025*. ([World Health Organization](#))
5. United Nations Environment Programme (UNEP). ([UNEP Food Waste Index Report 2024](#))
6. Benassai-Dalmau, R., et al. (2025). *Unequal Journeys to Food Markets: Continental-Scale Evidence from Open Data in Africa*. ([arXiv](#))